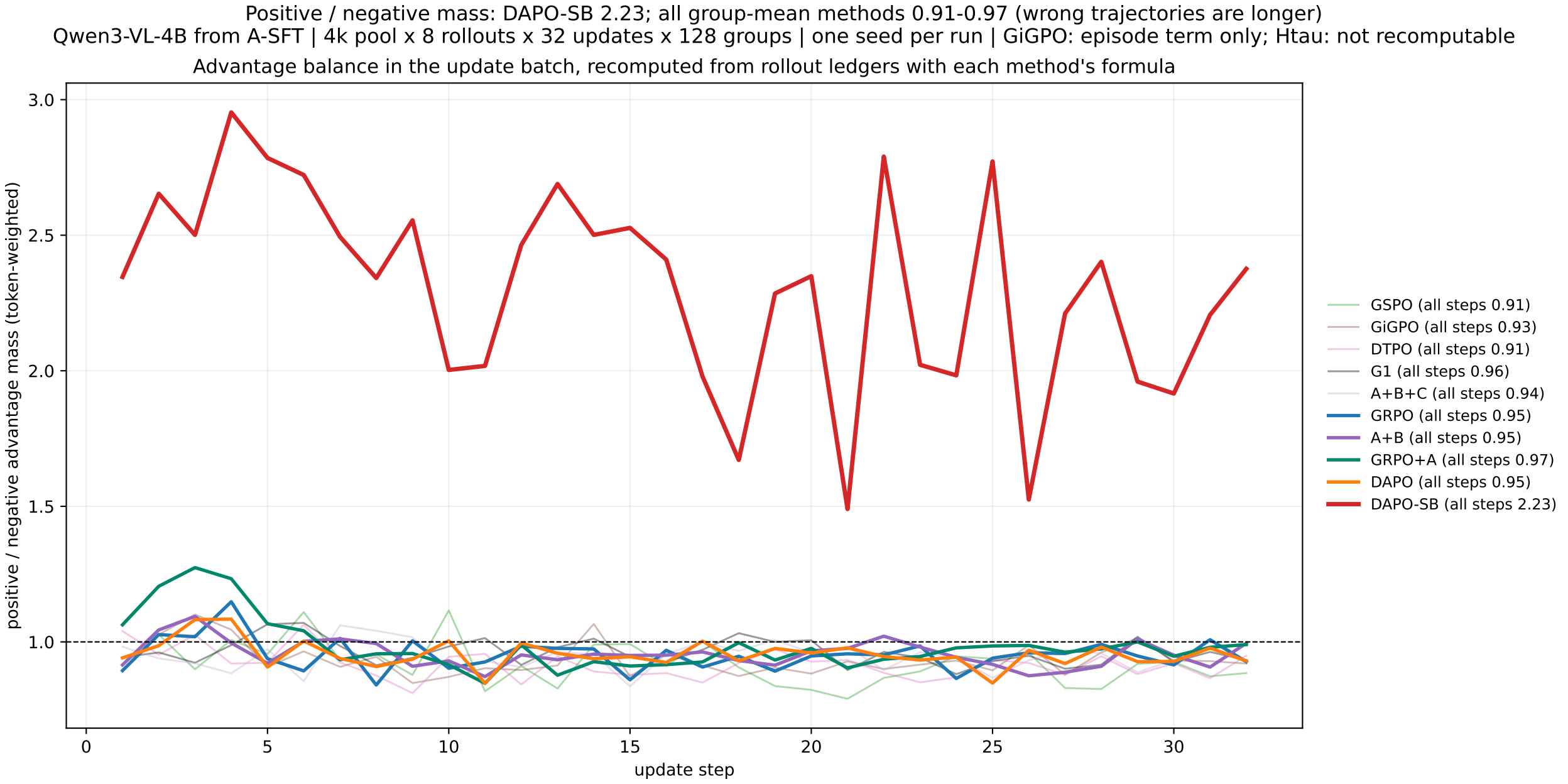
Positive / negative mass: DAPO-SB 2.23; all group-mean methods 0.91-0.97 (wrong trajectories are longer)
Qwen3-VL-4B from A-SFT | 4k pool x 8 rollouts x 32 updates x 128 groups | one seed per run | GiGPO: episode term only; Htau: not recomputable
Advantage balance in the update batch, recomputed from rollout ledgers with each method's formula
positive / negative advantage mass (token-weighted)
3.0
2.5
2.0
1.5
1.0
0
5
10
15
20
25
30
update step
GSPO (all steps 0.91)
GiGPO (all steps 0.93)
DTPO (all steps 0.91)
G1 (all steps 0.96)
A+B+C (all steps 0.94)
GRPO (all steps 0.95)
A+B (all steps 0.95)
GRPO+A (all steps 0.97)
DAPO (all steps 0.95)
DAPO-SB (all steps 2.23)